MATH848G
Index Theory on Manifolds



Taught by Jonathan Rosenberg
Notes taken by Haoran Li
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Department of Mathematics
University of Maryland

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1 Pseudodifferential calculus

Definition 1.1 A differential operator P of order m is of the form $p(x,D)=\sum_{|\alpha|\leq m} a_{\alpha(x)}D^{\alpha}$, where $a_{\alpha(x)}$ are functions and $D_j \coloneqq \frac{1}{i}\partial_j$, the symbol $p(x,\xi)$ associated with it is defined as $p(x,\xi)=\sum_{|\alpha|\leq m} a_{\alpha(x)}\xi^{\alpha}$, then we necessarily get the estimate, the principal symbol of $p(x,\xi)$ is $\sigma(P)(x,\xi)=\sum_{|\alpha|=m} a_{\alpha(x)}\xi^{\alpha}$.

If a differential operator P of order m is acting on functions on $\mathbb{C}^n \approxeq \mathbb{R}^{2n}$, then we may consider it as the form $p(x,D)=\sum_{|\alpha+\beta|\leq m} a_{\alpha,\beta}(x)D^{\alpha}\overline{D}^{\beta}$, and $D_j \coloneqq \frac{\partial}{\partial z_j}$, $\overline{D}_j \coloneqq \frac{\partial}{\partial \overline{z}_j}$, the symbol $p(x,\xi)$ associated with it is defined as $p(x,\xi)=\sum_{|\alpha+\beta|\leq m} a_{\alpha,\beta}(x)\xi^{\alpha}\overline{\xi}^{\beta}$.

Definition 1.2 On a manifold M, a differential operator of order m is a linear map $P: C^{\infty}(M)\to C^{\infty}(M)$ which in local charts has the above form.

More generally, let $E\to M, F\to M$ be vector bundles, a differential operator $P:\Gamma(E)\to\Gamma(F)$ of order m is a linear operator of the above form in local trivializations.

Notice $C^{\infty}(M)=\Gamma\left(\bigwedge^0 T^*M\right)$, another example is exterior derivative $d:\Gamma\left(\bigwedge^k T^*M\right)\to\Gamma\left(\bigwedge^{k+1} T^*M\right)$ which is a first order differential operator.

We can generalize the notion of differential operators (DO) to pseudodifferential operators (ψ DO) via symbols.

Definition 1.3 Let $\Omega\subseteq\mathbb{R}^n$ be an open set, the symbols of order m form the space

$$S^m(\Omega)\coloneqq\left\{f(x,\xi)\in C^{\infty}(\Omega\times\mathbb{R}^n)\;\middle|\; \begin{array}{c}\forall\alpha,\beta,K\Subset\Omega,\exists C_{K,\alpha,\beta},\\ \left|D_x^{\alpha}D_{\xi}^{\beta}f(x,\xi)\right|\leq C_{K,\alpha,\beta}(1+|\xi|)^{m-|\beta|}\end{array}\right\}$$

Given symbol $p(x,\xi)\in S^m(\Omega)$, it defines a pseudodifferential operator $p(x,D)$ of order m, such that $(p(x,D)f)(x)=\frac{1}{(2\pi)^n}\int p(x,\xi)\hat{f}(\xi)e^{ix\cdot\xi}\,\mathrm{d}\xi$.

All pseudodifferential operators of order m are defined this way, denoted as $\Psi^m(\Omega)$.

Remark 1.4 It might be more natural to think of $f\in C^{\infty}(\Omega\times(\mathbb{R}^n)^*)$ in the manifold setting.

Theorem 1.5 Let Ω,Ω' be open subsets of $\mathbb{R}^n$, $\Phi:\Omega\to\Omega'$ is a diffeomorphism, induces a bijection of $S^m(\Omega)\to S^m(\Omega')$.

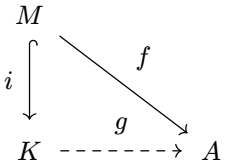
Proof. Since $\Phi:\Omega\to\Omega'$ is a diffeomorphism, $(\Phi_*)_x: T_x\Omega\to T_{\Phi(x)}\Omega'$ and $(\Phi^*)_x: T^*_{\Phi(x)}\Omega'\to T^*_x\Omega$ are isomorphisms. □

Corollary 1.6 Let M be a smooth manifold, $S^m(M)\subseteq\Gamma(T^*M)$ is well-defined.

Definition 1.7 The principal symbol of symbol $p(x,\xi)$ of order m is defined as $\lim_{|\xi|\to\infty}\frac{p(x,\xi)}{|\xi|^m}$.

2 K-theory

Definition 2.1 For commutative monoid M: there exists an abelian group K and $i : M \to K$ having the universal property

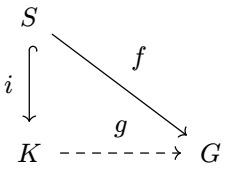


where A is an Abelian group.

Concrete construction: consider the set of formal differences $M \times M \cong \{m - n \mid m, n \in M\}$ with addition $(m_1 - n_1) + (m_2 - n_2) \coloneqq [(m_1 + m_2) - (n_1 + n_2)]$ which is an Abelian group with $0 - 0$ the identity and $n - m$ the inverse to $m - n$, let $K \coloneqq M \times M/\sim$, $m_1 - n_1 \sim m_2 - n_2$ if $m_1 + n_2 + m = m_2 + n_1 + m$ for some $m \in M$ (adding m to make sure it is an equivalence relation), and $i : M \to K, m \mapsto m - 0$.

Alternatively, consider the free Abelian group $F(M)$ generated by M, let $K \coloneqq F(M)/\sim$, $m +' n \sim (m + n)$, or mod the subgroup generated by $m +' n -' (m + n)$, here $+', -'$ are operations in $F(M)$.

More generally, for a semigroup S, there exists a group K and $i : S \to K$ having the universal property



where G is a group.

Concrete construction: consider the free group $F(S)$ generated by S, let $K \coloneqq F(S)/\sim$, $m *' n \sim (m * n)$, or mod the subgroup generated by $m *' n *' (m * n)^{-1}$, here $*', -^{-1}$ are operations in $F(M)$.

Definition 2.2 (Alternative definition of Grothendieck group) Let R be a finite dimensional k algebra (or more generally an Artinian ring), define Grothendieck group $G_0(R)$ to be the set of all finitely generated R modules mod relation, if $0 \to A \to B \to C \to 0$ is exact, then $[B] = [A] + [C]$, namely, $G_0(R) \coloneqq \{[M]\}$, where $[M]$ are equivalent classes of finitely generated R modules.

Definition 2.3 Two vector bundles $E \to X$, $F \to X$ are stably isomorphic if $E \oplus \varepsilon^n \approx F \oplus \varepsilon^n$, denoted as $E \underset{s}{\approx} F$, we also denote $E \underset{s}{\lesssim} F$ if $E \oplus \varepsilon^n \approx F \oplus \varepsilon^m$ for some n, m.

Remark 2.4 Here stably isomorphic does not imply isomorphic, for example, $TS^2 \underset{s}{\approx} \varepsilon^2$, since $\varepsilon^3 \approx T^2 \oplus NS^2 \approx T^2 \oplus \varepsilon^1$ whereas TS^2 is not trivial by the hairy ball theorem, and $NS^2 \approx \varepsilon^1$ is trivial because it is very easy to find a nonvanishing global section.

Definition 2.5 Let X be a topological space, then the Abelian group $\tilde{K}_{\mathbb{F}}(X)$ is defined to be the $\sim$-equivalent classes of $\mathbb{F}$-vector bundles over X, since all vector bundles over X with $\oplus$ form a commutative monoid, we can define its Grothendieck group $K(X)$ to be the K-theory, as it turns out, $K(X)$ also has a commutative ring structure with $\otimes$,

$$(E_1 - F_1) \otimes (E_2 - F_2) \coloneqq E_1 \otimes E_2 \oplus F_1 \otimes F_2 - E_1 \otimes F_2 \oplus F_1 \otimes E_2.$$

$K_{\mathbb{F}}(X, Y)$ is defined to be $\{E - F \in K_{\mathbb{F}}(X) \mid \exists \text{ isomorphism } \ E_Y \to F_Y\}$.

Note suppose X is compact Hausdorff, we can redefine the equivalence relation for the formal differences to be $E_1 - F_1 = E_2 - F_2$ if $E_1 \oplus F_2 \underset{s}{\approx} E_2 \oplus F_1$, because if $E_1 \oplus F_2 \oplus E \approx E_2 \oplus F_1 \oplus E$, for some E, then there exists E' such that $E \oplus E' \approx \varepsilon^m$ for some m, and any $E - F = (E \oplus F') - \varepsilon^m$ for some F' such that $F \oplus F' \approx \varepsilon^m$, thus we can define a map $K(X) \to \tilde{K}(X), E - \varepsilon^m \mapsto E$ which is obviously surjective, consider $K(X) \to K(*) \cong \mathbb{Z}$ by restricting a vector bundle to a point, and $K(*) \to K(X)$ by extending a vector space into a trivial vector bundle, we then get an exact sequence which splits

$$0 \longrightarrow K(*) \underset{\longleftarrow}{\longrightarrow} K(X) \longrightarrow \tilde{K}(X) \longrightarrow 0$$

Thus $K(X) \cong \tilde{K}(X) \oplus K(*) \cong \tilde{K}(X) \oplus \mathbb{Z}$.

Suppose X is a locally compact Hausdorff space, think of its one point compactification X^* as a pointed space, then $K(X) \cong \ker[K(X^*) \to K(*)] \cong \ker \tilde{K}(X^*) \cong K(X, *)$.

From this we have excision, if $Y \subseteq X$ are both compact Hausdorff, $\tilde{K}(X/Y) \cong \tilde{K}((X - Y)^*) \cong K(X - Y) \cong K(X, Y)$.

Remark 2.6 Normally $K(X)$ denotes $K_{\mathbb{C}}(X)$, $\mathrm{KO}(X)$ denotes $K_{\mathbb{R}}(X)$.

Theorem 2.7 K-theory forms an extraordinary cohomology theory.

Proof. □

Definition 2.8 Let X be a locally compact, Hausdorff space, consider all the complexes

$$\cdots \overset{\alpha_{-2}}{\to} E_{-1} \overset{\alpha_{-1}}{\to} E_0 \overset{\alpha_0}{\to} E_1 \overset{\alpha_1}{\to} E_2 \overset{\alpha_2}{\to} \cdots$$

with only finitely many $E_j \neq 0$, and exact off a compact set, give a semigroup structure by direct sum, then define $K(X) :=$ all such complexes$/\sim$, where a complex is equivalent to the identity if it is chain homotopic to an exact sequence off a compact set, $K(X, A)$ means an extra condition that the complexes in consideration are exact over A.

Proposition 2.9 The above two definitions of $K(X, A)$ coincide.

Proof. If X is compact, then every complex of vector bundles over X will be exact off a compact set and homotopic to a complex with zero maps, then define Euler characteristic map

$$K(X) \to K(X),$$

$$\left[0 \to E_1 \overset{\alpha_1}{\to} \cdots \to E_n \overset{\alpha_n}{\to} 0\right]$$

$$= \left[0 \to E_1 \overset{0}{\to} \cdots \to E_n \overset{0}{\to} 0\right] \mapsto \sum_{k=0}^{n} (-1)^k [E_i]$$

which would give an isomorphism.

If X is locally compact and Hausdorff, $K(X) \cong \ker[K(X^*) \to K(*)] \cong K(X)$. □

Theorem 2.10 Let X be a connected compact oriented two dimensional manifold without boundary (oriented closed surface), every complex vector bundle E is a sum of line bundles $[L_1] + \cdots + [L_n] - [L_{n+1}] - \cdots - [L_{n+m}]$, and line bundles are classified by Chern classes in $H^2(X, \mathbb{Z}) \cong \mathbb{Z}$, the rank and degree of E are defined as $\operatorname{rank} E = n - m$, $\deg E = \sum_{i \leq n} \deg[L_i] - \sum_{i > n} \deg[L_i]$. Therefore, $K(X) \overset{(\operatorname{rank}, \deg)}{\to} \mathbb{Z} \oplus \mathbb{Z}$ is an isomorphism.

3 Atiyah-Singer index theorem

Theorem 3.1 (Atiyah-Singer index theorem) M is a compact manifold, $p : T^*M \to M$ is the cotangent bundle, E, F are vector bundles over M, $P : \Gamma(E) \to \Gamma(F)$ is an elliptic ψDO of order m with principal symbol $\sigma(P) : p^*E \to p^*F$, P defines a Fredholm operator $H^s(E) \to H^{s-m}(F)$, from this we get its index (analytic index) $\mathrm{ind}_{a(P)}$, then

(1): $\mathrm{ind}_{a(P)}$ is independent of the choice of s.

(2): $\sigma(P)$ defines an element of $K(T^*M)$.

(3): $\mathrm{ind}_{a(P)}$ only depends on $[\sigma(P)] \in K(T^*M)$.

(4): $\exists$ homomorphism ind_t (topological index) : $K(T^*M) \to \mathbb{Z}$ independent of P, E, F such that $\mathrm{ind}_{a(P)} = \mathrm{ind}_{t([\sigma(P)])}$.

Proof. **(1):** Follows from what we did about ψDOs.

(2): Since $\sigma(P)$ is an elliptic operator, at $(x, \xi) \in T^*M$, $\sigma(P)(x, \xi)$ is a linear map $(p^*E)_{x,\xi} \to (p^*F)_{x,\xi}$ which is invertible except for ξ small, thus $0 \to p^*E \overset{\sigma(P)}{\to} p^*F \to 0$ is exact off a compact set, thus $\left[0 \to p^*E \overset{\sigma(P)}{\to} p^*F \to 0\right]$ defines an element in $K(T^*M)$.

(3): The K-theory class of $\sigma(P)$ only depends on $\sigma(P)$ over S^*M, which is compact, then you can extend it to T^*M just by homogeneity, $\sigma(P)(x, r\xi) = r^m\sigma(P)(x, \xi)$, varying this symbol over S^*M continuously, keeping it invertible, gives a homotopy of Fredholm operators, the index doesn't change since the index is locally constant, and such a homotopy preserves the K -theory class.

(4): □

Theorem 3.2 (Dolbeault theorem) Consider complex

$$\Omega^{0,0}(E) \overset{\overline{\partial}}{\to} \Omega^{0,1}(E) \overset{\overline{\partial}}{\to} \Omega^{0,2}(E) \overset{\overline{\partial}}{\to} \cdots$$

The j-th cohomology will be $H^j(M, \mathcal{O}_E)$.

Proof.

$$0 \to \mathcal{O}(E) \hookrightarrow \Omega^{0,0}(E) \overset{\overline{\partial}}{\to} \Omega^{0,1}(E) \overset{\overline{\partial}}{\to} \Omega^{0,2}(E) \overset{\overline{\partial}}{\to} \cdots$$

is a resolution of $\mathcal{O}(E)$ by fine sheaves as defined in Definition 3.3. □

Definition 3.3 A fine sheaf $\mathcal{F}$ over X is one with "partitions of unity"; more precisely for any open cover of the space X we can find a family of homomorphisms from the sheaf to itself with sum 1 such that each homomorphism is 0 outside some element of the open cover.

Example 3.4 M is compact Riemann surface, $E \to M$ is a holomorphic vector bundle, E is equipped with $\overline{\partial}$ operator, $\overline{\partial} : \Gamma(E) \to \Gamma(E \otimes \Omega^{0,1})$, $\overline{\partial} f \coloneqq \frac{\partial f}{\partial \overline{z}} \mathrm{d}\overline{z}$ (notice when E is trivial this is the normal $\overline{\partial}$ operator), $\overline{\partial}$ is an elliptic operator since its principal symbol is given by $\sigma\left(\overline{\partial}\right)(x, \xi) = \overline{\xi}$, thus $\mathrm{ind}_{a(\overline{\partial})} = \dim H^0(M, E) - \dim H^1(M, E)$, $H^0(M, E)$ is space of all holomorphic sections of E.

Lemma 3.5 $E \overset{p}{\to} X$ is a vector bundle, then the pullback bundle $p^*(E) \coloneqq E \times_X X$ is a trivial bundle.

Proof. □

Definition 3.6 Let X be a compact G space (meaning G acts on X), where G is a compact Lie group, a G-vector bundle E is a vector bundle over X which is also a G space, with $g : E_x \to E_{gx}$ a linear map on fibers. In particular, if X is a point, then a G-bundle is just a finite dimensional

representation of G, and $K_{G(X)}$ is the K-theory of such vector bundles, thus $R(G) \coloneqq K_{G(*)}$ is the representation ring of G, if X is a trivial G space, then a G-vector bundle is just a continuous family of G representations.

Example 3.7 (Examples of representation rings $R(G)$) $R(G) = \mathbb{Z}$.

Theorem 3.8 If X is a trivial G space, $K_{G(X)} \cong K(X) \otimes_{\mathbb{Z}} R(G)$.

If X is a free G space, $K_{G(X)} \cong K(X/G)$.

Theorem 3.9 (Equivariant Bott periodicity theorem) Let X be a locally compact G space, where G is a compact Lie group, V is a finite dimensional representation of G, then there is an isomorphism $K_G \to K_{G(V \times X)}$ defined as follows.

Consider the complex

$$0 \to \mathbb{C} = \overset{0}{\bigwedge} V \overset{\lambda_v}{\to} V = \overset{1}{\bigwedge} V \overset{\lambda_v}{\to} \overset{2}{\bigwedge} V \overset{\lambda_v}{\to} \cdots \overset{\lambda_v}{\to} \overset{n}{\bigwedge} V \to 0$$

where $\lambda_v \coloneqq v \wedge -$.

Theorem 3.10 (Thom isomorphism theorem in K-theory) Let X be a compact Hausdorff space, $E \to X$ is a complex vector bundle, then multiplication by λ_{E^*} gives an isomorphism $K(X) \to K(E)$.

Proof. Give an inner product on each fiber varying smoothly, $n = \dim E$, let $G = U(n)$, Y be the fiber bundle over X with fiber over x the Stiefel manifold of E_x, then G acts on Y and $\mathbb{C}^n \times Y$ freely, thus

$$K(X) = K(Y/G) \cong K_{G(Y)} \overset{\cong}{\to} K_{G(\mathbb{C}^n \times Y)}$$

$$\cong K((\mathbb{C}^n \times Y)/G) = K(E).$$

Here λ_E is the complex $\bigwedge(p^*E)$. □

4 Clifford algebra

Definition 4.1 Let $V = \mathbb{F}^n$ be a vector space with a quadratic form q, TV be its tensor algebra, let I be the ideal generated by elements of the form $v \otimes v + q(v, v)1$, the Clifford algebra is defined to be $Cl_n = \mathrm{Cliff}(V, q) = TV/I$, in particular, we only consider the case when $V = \mathbb{R}^n$ is the Euclidean space with the standard inner product, if $e_1, \cdots, e_n$ are the standard basis, $e_i \otimes e_i + 1 \in I$, $\left(e_i + e_j\right) \otimes \left(e_i + e_j\right) + 2 = e_i \otimes e_i + e_j \otimes e_j + e_i \otimes e_j + e_j \otimes e_i + 2 \in I$, we have $e_i^2 = -1$ and $e_i e_j = -e_j e_i, i \neq j$, thus $\dim Cl_n = 2^n$, $Cl_n = Cl_n^{\text{even}} \oplus Cl_n^{\text{odd}}$, $\dim Cl_n^{\text{even}} = \dim Cl_n^{\text{odd}} = 2^{n-1}$, Cl_n^{even} is a module over Cl_n^{even}.

Example 4.2 If $n = 1$, $Cl_1 \cong \mathbb{R}[e_1]/(e_1^2 + 1) \cong \mathbb{C}$.

If $n = 2$, $Cl_2 \cong \mathbb{R}[e_1, e_2]/(e_1^2 = e_2^2 = -1, e_1 e_2 = -e_2 e_1) \cong \mathbb{H}$.

Proposition 4.3 $Cl_{n+m} \cong Cl_n \otimes Cl_m$, here the tensor product is the graded tensor product.

Proof. □

Proposition 4.4 $Cl_n \otimes_{\mathbb{R}} \mathbb{C}$ is semisimple over $\mathbb{C}$ of dimension 2^n, and

$$Cl_n \otimes_{\mathbb{R}} \mathbb{C} = \begin{cases} M_{2^{\frac{n}{2}}}(\mathbb{C}) & \text{if } n \text{ is even} \\ M_{2^{\lfloor \frac{n}{2} \rfloor}}(\mathbb{C}) \oplus M_{2^{\lfloor \frac{n}{2} \rfloor}}(\mathbb{C}) & \text{if } n \text{ is odd} \end{cases}$$

Proof. □

5 Spin group

Definition 5.1 $Spin(n)$ is a subgroup of Cl_n^{even} consisting of elements of norm 1.

Example 5.2 $Spin(2) = \mathbb{T}$ is the circle group. Since $Cl_3^{\text{even}} \cong Cl_2 \cong \mathbb{H}$, $SU(2) \cong Spin(3)$ are the corresponding subgroups.

Definition 5.3 $Spin(n)$ naturally acts on $\mathbb{R}^n$ by $g \cdot v = gvg^{-1}$, then it is easy to see that $Spin(n) \to SO(n)$ is a double covering, if M is a Riemannian manifold, it has a principal bundle

$$\begin{array}{ccc} SO(n) & \longrightarrow & P \\ & & \downarrow \\ & & M \end{array}$$

where P_m are the oriented orthonormal frames on $T_m M$.

A spin structure on M is a lifting of P to a principal bundle for $Spin(n)$, note this may or may not exist.

Example 5.4 $\mathbb{C}P^{2n}$ doesn't have a spin structure.

Proposition 5.5 If spin structures exist, the set of such structures is acted on transitively by $H^1(M, \mathbb{Z}/2\mathbb{Z})$.

Proof. Consider the long exact sequence

$$H^1(M, \mathbb{Z}/2\mathbb{Z}) \to H^1(M, Spin(n)) \to H^1(M, SO(n)) \xrightarrow{w_2} H^2(M, \mathbb{Z}/2\mathbb{Z}).$$

□

Proposition 5.6 If n is even and M has a spin structure, fix it, then since $Spin(n) \subseteq Cl_n^{\text{even}} \subseteq Cl_n \subseteq Cl_n \otimes \mathbb{C} \cong M^{2^{\frac{n}{2}}}(\mathbb{C})$, and $Cl_n \otimes \mathbb{C}$ has a unique irreducible representation of dimension $2^{\frac{n}{2}}$, called the spin representation Δ, so we get a (complex) spin vector bundle $S = P_{Spin(n)} \times_{Spin(n)} \Delta$.

Definition 5.7 There is an elliptic operator acting on S via $\sum_{j=1}^n e_j \cdot \nabla e_j$, where $e_1, \cdots, e_n$ is a local orthonormal frame, the symbol is $\sum e_j \xi_j$, this is called the Dirac operator $\not{D}$.

Theorem 5.8 Assume M is a closed Riemannian spin manifold of dimension $n = 2l$, $ind\, \not{D} = \langle \hat{A}(M), [M] \rangle$, here $\hat{A}(M)$ is a polynomial in Pontryagin classes defined as follows: suppose $TM \otimes \mathbb{C} \cong L_1 \oplus \cdots \oplus L_l$ where L_j are complex line bundles with Chern class x_j, then $\hat{A}(M) = \prod_j \frac{x_j}{2\sinh\left(\frac{x_j}{2}\right)}$.

6 Homeworks

6.1 Homework 1

1.

Consider Hilbert space $(\mathcal{H}_s, \langle , \rangle)$, where

$$\mathcal{H}_s = \left\{ (c_m)_{m\in\mathbb{Z}^m} \;\middle|\; \sum_{m\in\mathbb{Z}^n} |c_m|^2 (1+|m|^2)^s < \infty \right\},$$

with the obvious vector space structure and $\langle (c_m), (d_m) \rangle = \sum_{m\in\mathbb{Z}^n} c_m \overline{d_m} (1+|m|^2)^s$, then Fourier transform $\mathcal{F} : H^s(\mathbb{T}^n) \to \mathcal{H}_s$, $f \mapsto \left(\hat{f}(m)\right)$ is an isometry, thus it suffices to prove $\mathcal{H}_{s_2} \Rightarrow \mathcal{H}_{s_1}$ is compact.

Suppose $\left\{(c_{\alpha m}) \in \mathcal{H}_{s_2}\right\}$ is a bounded set, namely, $\sum_{m\in\mathbb{Z}^n} |c_{\alpha m}|^2 (1+|m|^2)^{s_2} \le M$ for some M, $\forall k \in \mathbb{N}$, there exists N_k, for $|m| \ge N_k$, we have $(1+|m|^2)^{s_1-s_2} < \frac{1}{M2^{k+2}}$, and for $|m| \le N_k$, $(c_{\alpha m})_{|m|\le N_k}$ is a bounded set, thus having a convergent subsequence such that any $(c_{\alpha m}), (c_{\beta m})$ in that sequence satisfy $\sum_{|m|\le N_k} |c_{\alpha m} - c_{\beta m}|^2 (1+|m|^2)^{s_1} < \frac{1}{2^{k+1}}$, then by repeating this process, each time we get a subsequence out of the previous one, taking the diagonal sequence, we get a sequence (c_{km}) where for $|m| \ge N_k$, $(1+|m|^2)^{s_1-s_2} < \frac{1}{M2^{k+2}}$, and $\forall p, q \ge k$, $\sum_{|m|\le N_k} |c_{pm} - c_{qm}|^2 (1+|m|^2)^{s_1} < \frac{1}{2^{k+1}}$, then

$$\sum_{m\in\mathbb{Z}^n} |c_{pm} - c_{qm}|^2 (1+|m|^2)^{s_1}$$

$$\le \sum_{|m|\le N_k} |c_{pm} - c_{qm}|^2 (1+|m|^2)^{s_1} + \sum_{|m|\ge N_k} |c_{pm} - c_{qm}|^2 (1+|m|^2)^{s_2} (1+|m|^2)^{s_1-s_2}$$

$$< \frac{1}{2^{k+1}} + \frac{1}{M2^{k+2}} \sum_{m\in\mathbb{Z}^n} |c_{pm} - c_{qm}|^2 (1+|m|^2)^{s_2} < \frac{1}{2^k}.$$

Therefore (c_{km}) is a convergent sequence in $\mathcal{H}_{s_1}$.

2.

If θ is an irrational number and α is a rational number, then $m_1 + \theta m_2 + \alpha \ne 0$, $\forall (m_1, m_2) \in \mathbb{Z}^2$.

Since $\hat{f}(m) = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} f(z) z^{-m} \frac{\mathrm{d}z}{z}$, hence

$$\widehat{D_j f}(m) = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} z_j \frac{\partial}{\partial z_j} f(z) z^{-m-1} \,\mathrm{d}z$$

$$= \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} \left[\frac{\partial}{\partial z_j} f(z) z_j z^{-m-1} - f(z) \frac{\partial}{\partial z_j} \left(z_j z^{-m-1} \right) \right] \mathrm{d}z$$

$$= \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} \frac{\partial}{\partial z_j} f(z) z_j z^{-m-1} \,\mathrm{d}z - \frac{m_j}{(2\pi)^n} \int_{\mathbb{T}^n} f(z) z^{-m-1} \,\mathrm{d}z = m_j \hat{f}(m),$$

hence by Plancherel theorem

$$\left\| \hat{f}(m) \right\|^2 = \left\| \widehat{Df} \right\|^2 = \left\| \widehat{D_1 f} + \theta \widehat{D_2 f} + \alpha \hat{f}(m) \right\|^2$$

$$= \sum_{m\in\mathbb{Z}^n} \left| \widehat{D_1 f}(m) + \theta \widehat{D_2 f}(m) + \alpha \hat{f}(m) \right|^2$$

$$= \sum_{m\in\mathbb{Z}^n} \left| m_1 \hat{f} + \theta m_2 \hat{f} + \alpha \hat{f}(m) \right|^2$$

$$= \sum_{m\in\mathbb{Z}^n} |m_1+\theta m_2+\alpha|^2 \left|\hat{f}(m)\right|^2,$$

thus D is injective.

On the other hand, we want $\hat{f}(m)$ to be not rapidly decreasing while $(m_1+\theta m_2+\alpha)\hat{f}(m)$ is, consider Liouville number $\ell = \sum_{k=0}^{\infty} 2^{-k!}$ which is a transcendental number with the property that

$$\left|\ell - \sum_{k=0}^{n} 2^{-k!}\right| = \left|\sum_{k=n+1}^{\infty} 2^{-k!}\right| \leq 2^{1-(n+1)!} \leq \left(2^{-n!}\right)^n,$$

let $\frac{a_n}{b_n} \coloneqq \sum_{k=0}^{n} 2^{-k!}$, then a_n is an odd number whereas b_n is an even number with $\left|\ell - \frac{a_n}{b_n}\right| \leq \frac{1}{b_n^n}$, thus if we choose $\alpha = \frac{1}{2}, \theta = \ell, m_1 = \frac{a_n-1}{2}, m_2 = \frac{b_n}{2}$, then

$$|m_1+\theta m_2+\alpha| = |m_2|\left|\theta + \frac{2m_1+1}{2m_2}\right| \leq \frac{C}{|m_2|^{n-1}},$$

and as m_2 is big enough, we would have $\frac{1}{|m_2|} \leq \frac{C}{(1+|m|^2)^{\frac{1}{2}}}$, hence $|m_1+\theta m_2+\alpha| \leq \frac{C}{(1+|m|^2)^{\frac{n-1}{2}}}$, for such choices, make $\hat{f}(m) = 1$ and all others 0, then $\hat{f}(m)$ is not rapidly decreasing while $(m_1+\theta m_2+\alpha)\hat{f}(m)$ is.

6.2 Homework 2

1.

First we show that the n-th derivative of $p(x,\xi)=p(\xi)=\xi(1+\xi^2)^{-\frac{1}{2}}$ has the form $Q_{n+1}(\xi)(1+\xi^2)^{-\frac{2n+1}{2}}$ where $Q_{n+1}(\xi)$ is polynomial of ξ with degree no more than $n+1$, notice

$$\Big[Q_{n+1}(\xi)(1+\xi^2)^{-\frac{2n+1}{2}}\Big]'$$

$$=Q_{(n+1)'}(\xi)(1+\xi^2)^{-\frac{2n+1}{2}}-(2n+1)\xi Q_{n+1}(\xi)(1+\xi^2)^{-\frac{2n+3}{2}}$$

$$=\Big[(1+\xi^2)Q_{(n+1)'}(\xi)-(2n+1)\xi Q_{n+1}(\xi)\Big](1+\xi^2)^{-\frac{2(n+1)+1}{2}},$$

we can define $Q_1(\xi)=\xi$ and $Q_{n+2}(\xi)=(1+\xi^2)Q_{(n+1)'}(\xi)-(2n+1)\xi Q_{n+1}(\xi)$ inductively, then $\big|Q_{n+1}(\xi)\big|\le C^{n+1}_{n(1+|\xi|)}$ for some constant C_n only depends on n, and $(1+\xi^2)^{-\frac{1}{2}}\ge\sqrt{2}(1+|\xi|)^{-1}$, thus $\left|Q_{n+1}(\xi)(1+\xi^2)^{-\frac{2n+1}{2}}\right|\le\sqrt{2}C^{-n}_{n(1+|\xi|)}$, also $|p(x,\xi)|>\frac{1}{2}$ when ξ is sufficiently large, thus P is an elliptic pseudo-differential operator of order 0, the symbol for $I-P^2$ is $\frac{1}{1+\xi^2}$, it is not a smoothing operator since $\widehat{e^{-|x|}}(\xi)=\frac{2}{1+\xi^2}$ and apply Fourier inversion theorem then kernel corresponding to operator $I-P^2$ is

$$K(x,y)=\frac{1}{2\pi}\int_{\mathbb{R}}\frac{e^{i(x-y)\xi}}{1+\xi^2}\,\mathrm{d}\xi=\frac{1}{2}e^{-|x-y|}$$

which is not smooth along the diagonal.

P is not Fredholm on $L^2(\mathbb{R})$, because Fourier transform is an isometry on $L^2(\mathbb{R})$, P corresponds to multiplication by $\frac{\xi}{\sqrt{1+\xi^2}}$ on $L^2(\mathbb{R})$, it is easy to see $\ker P=0$, as for cokernel of P, consider functions $\xi^\alpha\mathbb{1}_{[-1,1]}$ with $-\frac{1}{2}<\alpha\le\frac{1}{2}$, they belong to $L^2(\mathbb{R})$ but without preimages in $L^2(\mathbb{R})$, and any nonzero linear combination of them also doesn't have a preimage, so they are linearly independent representatives of cokernel of P, hence cokernel of infinite dimension and P is not Fredholm on $L^2(\mathbb{R})$.

Suppose f is a periodic function on $\mathbb{R}$ of period 2π, then

$$P(x,D)f=\int_{\mathbb{R}}\frac{\xi}{\sqrt{1+\xi^2}}\hat{f}(\xi)e^{ix\xi}\,\mathrm{d}\xi$$

$$=\int_{\mathbb{R}}\frac{\xi}{\sqrt{1+\xi^2}}e^{-i2\pi\xi}\hat{f}(\xi)e^{i(x+2\pi)\xi}\,\mathrm{d}\xi$$

$$=\int_{\mathbb{R}}\frac{\xi}{\sqrt{1+\xi^2}}\hat{f}(\xi)e^{i(x+2\pi)\xi}\,\mathrm{d}\xi=P(x+2\pi,D)f,$$

hence $P(x,D)f$ is also a periodic function on $\mathbb{R}$ of period 2π, thus we may define P as an operator on $S^1\cong\mathbb{R}/2\pi\mathbb{Z}$, notice

$$P(x,D)(e^{inx})=\int_{\mathbb{R}}\frac{\xi}{\sqrt{1+\xi^2}}\widehat{e^{inx}}(\xi)e^{ix\xi}\,\mathrm{d}\xi$$

$$=\int_{\mathbb{R}}\frac{\xi}{\sqrt{1+\xi^2}}\delta_{n(\xi)}e^{ix\xi}\,\mathrm{d}\xi=\frac{n}{\sqrt{1+n^2}}e^{inx},$$

$$P(x,D)\Big(\sum c_ne^{inx}\Big)=\sum\frac{nc_n}{\sqrt{1+n^2}}e^{inx}$$

then it is easy to see $\operatorname{ind}P=\dim\ker P-\dim\operatorname{coker}P=1-1=0$.

2.

After Fourier transform, P_K corresponds to multiplication by $K^2 - n^2$, suppose for a $\|f\|_{H^2(S^1)} = \sum (1 + |n|^2)^2 \left|\hat{f}(n)\right|^2 = 1$, we have

$$\left\| \left(P_{K_2} - P_{K_1}\right) f \right\|_{L^2(S^1)} = \sum \left| (K_2^2 - K_1^2) \hat{f}(n) \right|^2$$

$$= (K_2^2 - K_1^2)^2 \sum \left|\hat{f}(n)\right|^2 \leq (K_2^2 - K_1^2)^2,$$

thus P_K varies continuously with K, when $K = 0$, the periodic functions in the kernel of P_0 are only constants, i.e. $\dim \ker P_0 = 1$, on the other hand, consider linear operator $T : L^2(S^1) \to \mathbb{R}, f \mapsto \int_0^{2\pi} f \, \mathrm{d}x$, notice $\ker T = \left\{ f \in L^2(S^1) \;\middle|\; \int_0^{2\pi} f \, \mathrm{d}x = 0 \right\} = \operatorname{im} P_0$ since for such f, periodic function $\int_0^x \int_0^y f(z) \, \mathrm{d}z \, \mathrm{d}y - \int_0^{2\pi} z f(z) \, \mathrm{d}z$ is a preimage, hence cokernel of P_0 is of dimension 1, thus $\operatorname{ind} P = \dim \ker P - \dim \operatorname{coker} P = 1 - 1 = 0$.

More directly, $\sum \frac{\frac{1}{2\pi} \int_0^{2\pi} f(y) e^{-iny} \, \mathrm{d}y}{K^2 - n^2} e^{inx}$ will be a solution when K is not an integer.

6.3 Homework 3

1.

(i)

Since $d_0(f) = \mathrm{d}f = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y$,

$$d_1^*(f\,\mathrm{d}x \wedge \mathrm{d}y) = - \star\, d \star (f\,\mathrm{d}x \wedge \mathrm{d}y) = - \star \mathrm{d}f$$

$$= - \star \left(\frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y \right) = \frac{\partial f}{\partial y}\,\mathrm{d}x - \frac{\partial f}{\partial x}\,\mathrm{d}y,$$

the symbol (or in this case the principal symbol) of D is $\begin{pmatrix} i\xi_1 & i\xi_2 \\ i\xi_2 & -i\xi_1 \end{pmatrix} = i\begin{pmatrix} \xi_1 & \xi_2 \\ \xi_2 & -\xi_1 \end{pmatrix}$ is invertible off the zero section, hence D is elliptic.

Alternatively, $d_0(f) = \mathrm{d}f = \frac{\partial f}{\partial z}\,\mathrm{d}z + \frac{\partial f}{\partial \overline{z}}\,\mathrm{d}\overline{z}$, $d_1^*(f\,\mathrm{d}z \wedge \mathrm{d}\overline{z}) = i\frac{\partial f}{\partial z}\,\mathrm{d}z - i\frac{\partial f}{\partial \overline{z}}\,\mathrm{d}\overline{z}$, thus the symbol (principal symbol) of D would be $\begin{pmatrix} \xi & i\xi \\ \bar{\xi} & -i\bar{\xi} \end{pmatrix}$ which is invertible off the zero section, thus D is elliptic.

(ii)

(iii)

$D^* = d_0^* + d_1$, thus

$$D^*D = (d_0^* + d_1)(d_0 + d_1^*) = d_0^*d_0 + d_0^*d_1^* + d_1d_0 + d_1d_1^*$$

$$= d_0^*d_0 + (d_1d_0)^* + d_1d_0 + d_1d_1^* = d_0^*d_0 + d_1d_1^*,$$

then

$$\langle D(f+\omega), D(f+\omega)\rangle = \langle D^*D(f+\omega), (f+\omega)\rangle$$

$$= \langle d_0^*d_0 f, f\rangle + \langle d_1d_1^*\omega, \omega\rangle$$

$$= \langle d_0 f, d_0 f\rangle + \langle d_1^*\omega, d_1^*\omega\rangle,$$

thus $D(f+\omega) = 0 \Leftrightarrow d_0 f = d_1^*\omega = 0$,

$$\ker D = \ker d_0 \oplus \ker d_1^* \cong H^0_{dR}(S^2, \mathbb{R}) \oplus H^2_{dR}(S^2, \mathbb{R})$$

$$\cong H^0(S^2, \mathbb{R}) \oplus H^2(S^2, \mathbb{R}),$$

but the cokernel of D is just $H^1_{dR}(S^2, \mathbb{R}) \cong H^1(S^2, \mathbb{R}) = 0$, therefore $\operatorname{ind} D = \dim\ker D - \dim\operatorname{coker} D = 2$.

2.

6.4 Homework 4

1.

(i)

Notice $\ker \overline{\partial}_{\mathcal{O}(n)} = H^0(\mathbb{C}P^1, \mathcal{O}(n))$, and $\operatorname{coker} \overline{\partial}_{\mathcal{O}(n)} = 0$, G acts on $H^0(\mathbb{C}P^1, \mathcal{O}(n))$ by $(g \cdot f)(z) = f(g^{-1}z)$ where $f(z)$ is a homogeneous polynomial of z_0, z_1 of degree n, suppose $f = \sum_{k=0}^{n} \alpha_k z_0^k z_1^{n-k}$, with $\alpha_{k_0} \neq 0$ for some k_0, consider $h_l = \operatorname{diag}(\overline{u_l}, u_l), h = \sum_{l=0}^{m} c_l h_l$ acting on f, we get

$$(h \cdot f)(z) = \sum_{l=0}^{m} (h_l \cdot f)(z) = \sum_{l=0}^{m} \sum_{k=0}^{n} c_l u_l^{n-2k} \alpha_k z_0^k z_1^{n-k}$$

$$= \sum_{k=0}^{n} \left(\sum_{l=0}^{m} c_l u_l^{n-2k} \right) \alpha_k z_0^k z_1^{n-k},$$

take distinct $u_1, \cdots, u_n$ with $|u_l| = 1$, and consider system of equations $\sum_{l=0}^{n} c_l u_l^{n-2k} = 0, \forall k \neq k_0$, and $\sum_{l=0}^{n} c_l u_l^{n-2k_0} = \frac{1}{\alpha_{k_0}}$ which has a solution of c_l's since determinant

$$\left|\left(u_l^{n-2k}\right)\right| = \left|\left(u_l^n (u_l^{-2})^k\right)\right| = \left(\prod |u_l|^n\right) \left|\left((u_l^{-2})^k\right)\right| \neq 0$$

and $\left(u_l^{-2}\right)^k$ is a Vandermonde matrix, thus $(h \cdot f)(z) = z_0^{k_0} z_1^{n-k_0}$, then $z_0^{k_0} z_1^{n-k_0} \in \mathbb{C}Gf$.

Let $g = \begin{pmatrix} u & w \\ -\overline{w} & \overline{u} \end{pmatrix}$, $g \cdot z_0^{k_0} z_1^{n-k_0} = (uz_0 + wz_1)^{k_0} (-\overline{w} z_0 + \overline{u} z_1)^{n-k_0}$ which in expansion has nonzero coefficient for term z_0^n, use the argument we have $z_0^n \in \mathbb{C}Gf$, then $g \cdot z_0^n = (uz_0 + wz_1)^n$ which in expansion has nonzero coefficients for each term $z_0^k z_1^{n-k}$, by the argument again, we know each term $z_0^k z_1^{n-k} \in \mathbb{C}Gf$, thus $\mathbb{C}Gf$ generates all homogeneous polynomials of degree n.

Therefore this representation is irreducible.

Notice that $\operatorname{diag}(\overline{z}, z)\left(z_0^k z_1^{n-k}\right) = z_0^k z_1^{n-k} z^{n-2k}$ thus this is a subrepresentation $\langle z_0^k z_1^{n-k} \rangle$ corresponds to t^{n-2k}, thus the restriction of this representation to H is the direct sum of $\langle z_0^k z_1^{n-k} \rangle$ which is $t^n + \cdots + t^{-n} = \chi_n$.

This question I got some hints from Artavia Josue.

(ii)

Obviously $\mathbb{Z}[u] \subseteq \mathbb{Z}\left[t, t^{-1}\right]^\sigma$, on the other hand, if $f(t) = \sum_{n=-\infty}^{\infty} c_n t^n \in \mathbb{Z}\left[t, t^{-1}\right]^\sigma$, then $\sum_{n=-\infty}^{\infty} c_n t^{-n} = \sigma f(t) = f(t) = \sum_{n=-\infty}^{\infty} c_n t^n$, which implies that $c - n = c_{-n}$, thus $f(t) = c_0 + \sum_{n=1}^{\infty} c_{n(t^n + t^{-n})}$, but note that

$$2(t + t^{-1})^n = \sum_{i=0}^{n} \binom{n}{i} t^{n-i} t^{-i} + \sum_{i=0}^{n} \binom{n}{i} t^i t^{i-n}$$

$$= \sum_{i=0}^{n} \binom{n}{i} t^{n-2i} + \sum_{i=0}^{n} \binom{n}{i} t^{2i-n}$$

$$= \sum_{i=0}^{n} \binom{n}{i} (t^{n-2i} + t^{2i-n}).$$

Thus $t^n + t^{-n} \in \mathbb{Z}[u]$, $\mathbb{Z}[u] = \mathbb{Z}\left[t, t^{-1}\right]^\sigma$.

Each power of u is a sum of characters of the form χ_n because $t^n + t^{-n} = \chi_{n+2} - \chi_n$.

Therefore the image of $R(G)$ in $R(H)$ is contained in $\mathbb{Z}[u]$ and the restriction map is injective, then consider $R(H) \to R(G) \to R(H), t^n \mapsto V_n \mapsto \chi_n$, thus $\mathbb{Z}[u]$ will be contained in the image, hence $R(H) \to R(G)$ is surjective and $R(G) \to R(H)$ has precisely image $\mathbb{Z}[u]$.

By Serre duality, we know that $\dim H^1(\mathbb{C}P^1, \mathcal{O}(n)) = \dim H^0(\mathbb{C}P^1, K \otimes \mathcal{O}(-n)) = \dim H^0(\mathbb{C}P^1, \mathcal{O}(-n-2))$, when $n = -1$, t^{-1} is mapped to 0, when $n \leq -2$, the image of t^n is $-V_{-n-2}$, which restricts to χ_{-n-2}.

2.

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